# E58: Sharp Bound for the Band-Selective Green Multiplier

Richard J. Reyes
Independent Researcher

October 26, 2026

**Abstract**

For the fourth-order band-selective operator $\mathcal{L} = r + a(\Delta + k_\star^2)^2$ on $\mathbb{R}^d$, with $r > 0$, $a \geq 0$, and $k_\star \geq 0$, we prove the pointwise Fourier-multiplier bound $0 < G(k) \leq 1/r$. For $a > 0$, the bound is sharp exactly on the selected shell $|k| = k_\star$. We further prove the exact $L^2$ resolvent norm, identify the discrete-spectrum qualification, and distinguish positivity of the Fourier multiplier from positivity of the physical-space Green function.

## 1 Definitions and principal result

Fix an integer $d \geq 1$. On the Schwartz class $\mathcal{S}(\mathbb{R}^d)$, consider

$$\mathcal{L} := r + a(\Delta + k_\star^2)^2, \qquad r > 0, \quad a \geq 0, \quad k_\star \geq 0. \tag{1}$$

We use the Fourier-transform convention

$$\widehat{u}(k) = \int_{\mathbb{R}^d} e^{-ik\cdot x} u(x)\, dx, \qquad u(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{ik\cdot x} \widehat{u}(k)\, dk. \tag{2}$$

Write

$$\ell(k) := r + a\big(|k|^2 - k_\star^2\big)^2, \qquad G(k) := \frac{1}{\ell(k)}. \tag{3}$$

**Theorem 1** (E58: positive, sharply bounded Green multiplier)**.** *For every $k \in \mathbb{R}^d$,*

$$\boxed{0 < G(k) \leq \frac{1}{r}.} \tag{4}$$

*If $a > 0$, equality holds if and only if $|k| = k_\star$; in particular,*

$$\sup_{k\in\mathbb{R}^d} G(k) = \frac{1}{r}. \tag{5}$$

*If $a = 0$, $G(k) = 1/r$ for all $k$.*

*Proof.* Since $\widehat{\Delta u}(k) = -|k|^2 \widehat{u}(k)$,

$$\widehat{(\Delta + k_\star^2)u}(k) = (k_\star^2 - |k|^2)\widehat{u}(k),$$
$$\widehat{(\Delta + k_\star^2)^2 u}(k) = (|k|^2 - k_\star^2)^2 \widehat{u}(k).$$

Therefore $\widehat{\mathcal{L}u}(k) = \ell(k)\widehat{u}(k)$ with $\ell$ as in (3). For every $k$,

$$(|k|^2 - k_\star^2)^2 \geq 0, \qquad a(|k|^2 - k_\star^2)^2 \geq 0, \tag{6}$$

and hence

$$0 < r \leq r + a(|k|^2 - k_\star^2)^2 = \ell(k). \tag{7}$$

Taking reciprocals of positive numbers reverses the inequality:

$$0 < \frac{1}{\ell(k)} \leq \frac{1}{r}, \tag{8}$$

which proves (4).

For $a > 0$, equality holds precisely when $a(|k|^2 - k_\star^2)^2 = 0$, or equivalently when $|k|^2 = k_\star^2$. Since $k_\star \geq 0$, this is exactly $|k| = k_\star$. That set is nonempty in $\mathbb{R}^d$, even if $k_\star = 0$, proving (5). When $a = 0$, the symbol is identically $r$, so $G \equiv 1/r$. □

## 2 Inverse operator and exact $L^2$ estimate

**Proposition 1** (Sharp continuum resolvent bound)**.** *Let $\mathcal{L}$ be as in* (1)*. For each $f \in L^2(\mathbb{R}^d)$, there exists a unique $u \in L^2(\mathbb{R}^d)$ satisfying $\mathcal{L}u = f$ in the sense of tempered distributions. It is defined by*

$$\widehat{u}(k) = G(k)\widehat{f}(k). \tag{9}$$

*Moreover,*

$$\boxed{\|\mathcal{L}^{-1}\|_{L^2 \to L^2} = \frac{1}{r}.} \tag{10}$$

*If $a > 0$, then $u \in H^4(\mathbb{R}^d)$.*

*Proof.* By Theorem 1, $G \in L^\infty(\mathbb{R}^d)$ and $\|G\|_{L^\infty} \leq 1/r$. For any $f \in L^2$, the product $G\widehat{f} \in L^2$, so Plancherel's theorem defines the unique $u \in L^2$ with (9) and gives

$$\|u\|_2^2 = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} |G(k)|^2 |\widehat{f}(k)|^2 \, dk \leq \frac{1}{r^2} \|f\|_2^2. \tag{11}$$

Since $\ell(k)G(k) = 1$, equation (9) solves $\mathcal{L}u = f$ distributionally. Conversely, if $\mathcal{L}u = 0$ and $u \in L^2$, then $\ell\widehat{u} = 0$ and $\ell > 0$, hence $u = 0$.

To show that (11) is sharp, fix $\varepsilon > 0$ and let

$$E_\varepsilon := \left\{ k \in \mathbb{R}^d : ||k|^2 - k_\star^2| < \varepsilon \right\}. \tag{12}$$

This measurable set has positive, finite measure for every $\varepsilon > 0$ (including when $k_\star = 0$). Choose a nonzero $f_\varepsilon \in L^2$ whose Fourier transform is supported in $E_\varepsilon$. On this set,

$$G(k) \geq \frac{1}{r + a\varepsilon^2}, \tag{13}$$

so Plancherel implies

$$\frac{\|\mathcal{L}^{-1} f_\varepsilon\|_2}{\|f_\varepsilon\|_2} \geq \frac{1}{r + a\varepsilon^2}. \tag{14}$$

Letting $\varepsilon \downarrow 0$ gives $\|\mathcal{L}^{-1}\|_{L^2 \to L^2} \geq 1/r$, which, together with (11), proves (10).

Finally, if $a > 0$, the function $(1 + |k|^2)^2 G(k)$ is bounded: its denominator grows like $a|k|^4$ at high frequency and stays strictly positive at finite frequency. Hence

$$\int_{\mathbb{R}^d} (1 + |k|^2)^4 |\widehat{u}(k)|^2 \, dk \le C^2 \int_{\mathbb{R}^d} |\widehat{f}(k)|^2 \, dk < \infty, \tag{15}$$

so $u \in H^4(\mathbb{R}^d)$. □

**Remark 1** (Norm attainment)**.** *For $a > 0$, the equality set $\{k : |k| = k_\star\}$ has Lebesgue measure zero. Thus the $L^2 \to L^2$ operator norm in* (10) *is approached by concentrating Fourier mass near the shell, but is not attained by any nonzero $L^2$ input. For $a = 0$, every nonzero input attains it.*

## 3 Energy (coercivity) proof of the upper estimate

When $a > 0$ and $u \in H^4(\mathbb{R}^d)$, integration by parts, or Plancherel's theorem, yields

$$\begin{aligned} \langle u, \mathcal{L}u \rangle_{L^2} &= r\|u\|_2^2 + a\|(\Delta + k_\star^2)u\|_2^2 \\ &\ge r\|u\|_2^2. \end{aligned} \tag{16}$$

If $\mathcal{L}u = f$, then Cauchy–Schwarz gives

$$r\|u\|_2^2 \le \mathrm{Re}\langle u, f \rangle_{L^2} \le \|u\|_2 \|f\|_2. \tag{17}$$

Dividing by $\|u\|_2$ when it is nonzero proves $\|u\|_2 \le r^{-1}\|f\|_2$; when $u = 0$, the estimate is immediate. For $a = 0$, $\mathcal{L} = rI$ and the estimate is an identity.

## 4 Band selectivity and the discrete-spectrum caveat

Assume $a > 0$ and set $\delta := ||k|^2 - k_\star^2| \ge 0$. Then

$$G(\delta) = \frac{1}{r + a\delta^2}, \qquad \frac{dG}{d\delta} = -\frac{2a\delta}{(r + a\delta^2)^2} < 0 \quad (\delta > 0). \tag{18}$$

Thus the response is strictly largest on the shell and strictly decreases with detuning. At high detuning,

$$G(\delta) \sim \frac{1}{a\delta^2} \qquad (\delta \to \infty). \tag{19}$$

For a periodic box, the same Fourier-multiplier inequality holds at every admissible discrete wavevector $k \in \Lambda$. However,

$$\|\mathcal{L}^{-1}\|_{L^2 \to L^2} = \sup_{k \in \Lambda} G(k) \le \frac{1}{r}, \tag{20}$$

with equality for $a > 0$ if and only if the discrete spectrum contains a wavevector with $|k| = k_\star$. The maximum is smaller than $1/r$ otherwise.

## 5 Fourier multiplier versus physical-space Green function

The inequality $0 < G(k) \le 1/r$ is a *Fourier-space* statement. It does not imply that the inverse Fourier transform $g = \mathcal{F}^{-1}G$ is pointwise nonnegative, nor does it provide a pointwise bound $g(x) \le 1/r$.

To see why nonnegativity fails, consider $d = 1$, $k_\star = 0$, $r > 0$, $a > 0$. Write

$$b := \left(\frac{r}{a}\right)^{1/4}, \qquad t := \frac{b|x|}{\sqrt{2}}. \tag{21}$$

The explicit spatial Green function for $\mathcal{L} = r + a\, d^4/dx^4$ is

$$g(x) = \frac{1}{2\sqrt{2}\, a^{1/4} r^{3/4}} e^{-t}\big(\cos t + \sin t\big). \tag{22}$$

For completeness, the formula follows from the even, decaying solution to $ag'''' + rg = 0$ away from $x = 0$. Indeed, with $c = b/\sqrt{2}$, the function $h(x) = e^{-c|x|}(\cos(c|x|)+\sin(c|x|))$ satisfies $ah'''' + rh = 0$ for $x \neq 0$, is continuous with $h'(0^+) = h'(0^-) = 0$, and has $h'''(0^+) - h'''(0^-) = 8c^3$. Multiplying by $C = (8ac^3)^{-1} = (2\sqrt{2}\, a^{1/4} r^{3/4})^{-1}$ gives $ag'''' + rg = \delta_0$ in the sense of distributions. Since the Fourier transform of this identity yields $\widehat{g}(k) = 1/(r + ak^4)$, equation (22) is exactly the inverse Fourier transform of $G$ in this example.

At $t = \pi$, one has $\cos t + \sin t = -1$, so $g(x) < 0$, even though $G(k) > 0$ for every $k$. Consequently, positivity of the Fourier multiplier must not be confused with pointwise positivity of the spatial Green function.

## 6 Scope and conclusion

The assumptions $r > 0$ and $a \ge 0$ are essential to the unconditional bound. If $r = 0$ and $a > 0$, the Fourier symbol vanishes on the selected shell. If $a < 0$, it is not globally bounded below by $r$ and can vanish at real frequencies. The result establishes a bounded *linear* spectral response; it does not, by itself, establish nonlinear pattern stability, spatial localization, or physical wavelength selection.

$$\boxed{\forall k \in \mathbb{R}^d : \quad 0 < \frac{1}{r + a(|k|^2 - k_\star^2)^2} \le \frac{1}{r}, \qquad \|\mathcal{L}^{-1}\|_{L^2 \to L^2} = \frac{1}{r}.}$$